26900231 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEFSQ301 - COCONUT PROCESSING TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. State the change in husk colour used as a visual maturity index for coconuts suitable for commercial oil milling. [K] / [CO1]

   a). Bright translucent yellow color

   b). Complete transition from green to deep dark brown

   c). Uniform pale ivory white shade

   d). Mottled purple patches across the exocarp

2. Identify the purpose of mechanical drying. [K] / [CO1]

   a). Speed up dryin

   b). Increase moisture

   c). Reduce temperature

d). Increase spoilage

3. Identify the factore that largely influences the quality of copra. [U] / [CO1]

a). Moisture content

b). Size of seed

c). Dehusking process

d). Maturity of seed

4. Name the biological functionality associated with lauric acid in coconut kernel products. [K] / [CO1]

a). It converts directly into structural petroleum substitutes inside muscle tissue

b). It exhibits distinct antimicrobial and antiviral properties via monolaurin formation

c). It permanently blocks carbohydrate absorption in the human lower intestine

d). It acts as a heavy sedative agent affecting the central nervous system

5. Identify the purpose of ventilation in coconut storage. [U] / [CO1]

a). Removal of excess moisture

b). Increase in humidity

c). Promotion of fungal growth

d). Increase in temperature

6. Identify why he suitability of coconut cultivation in coastal regions is mainly due to: [U] / [CO1]

a). Dry climate

b). Sandy soils and humid conditions

c). Snowfall

d). Low temperatures

7. Indicate the suitable measure for reducing oxidative deterioration of coconut oil during storage: [U] / [CO2]

a). Exposure to sunlight

b). Storage in well-sealed containers

c). Addition of water

d). Storage at high temperature

8. Identify the condition most likely to improve oil recovery during expeller pressing: [U] / [CO2]

a). Inadequate pressure with improperly conditioned kernel

b). Appropriate pressure with properly prepared kernel

c). Excessive moisture with reduced pressing pressure

d). Uneven particle size with uncontrolled temperature

9. Select the purpose of grating coconut kernel before oil extraction: [U] / [CO2]

a). To increase the size of the kernel

b). To break the cellular structure and facilitate oil release

c). To remove coconut water

d). To increase the moisture content

10. Select the main importance of deodorization during coconut oil refining: [U] / [CO2]

a). Removal of phospholipids and gums

b). Removal of free fatty acids using alkali

c). Removal of undesirable odour and flavour compounds

d). Removal of colouring pigments

11. Name the traditional method used to separate oil from manually extracted coconut milk without machinery. [K] / [CO1]

a). By filtering the cold liquid through specialized multi-layered ceramic membranes

b). By boiling the milk directly in open pans to evaporate water and precipitate proteins

c). By injecting high-pressure compressed air to create a floating grease foam

d). By adding concentrated sulfuric acid to digest the carbohydrate fractions

12. Identify the statement which explains the scope of coconut processing technology. [U] / [CO1]

a). It is limited to coconut harvesting

b). It includes cultivation, processing, quality improvement, and product development

c). It focuses only on coconut oil extraction

d). It deals only with coconut export.

13. Select the main component responsible for the TSS of coconut water: [U] / [CO3]

a). Proteins

b). Sugars

c). Fats

d). Minerals

14. Select the quality attribute mainly assessed through sensory evaluation of coconut products: [U] / [CO3]

a). Moisture percentage

b). Acidity value

c). Ash content

d). Odour

15. Select the quality parameter related to the degree of unsaturation of coconut oil: [U] / [CO3]

a). Acid value

b). Iodine value

c). Peroxide value

d). Saponification value

16. Interpret the significance of tender coconut water during diarrhoea and vomiting. [U] / [CO4]

a). Potential to help prevent dehydration

b). Ability to increase gastrointestinal activity

c). High concentration of total sugars

d). Ability to replace all nutrient requirements

17. Select the characteristics differentiating coconut apple from tender coconut. [K] / [CO4]

a). Developmental stage of the coconut b). Colour of the outer husk alone

c). Shape of the coconut shell alone d). Size of the coconut tree

18. Identify the packaging property associated with reduced rancidity in coconut oil during prolonged storage: [K] / [CO4]

a). Transparency

b). Moisture-barrier property

c). Flexibility only

d). Low mechanical strength

19. Predict the particle size produced by the Devil Disintegrator [An] / [CO4]

a). 1-5mm b). 5-10mm c). 10-15mm d). 15-20mm

20. Identify the coconut component associated with antioxidant activity. [K] / [CO4]

a). Phenolic compounds

b). Starch

c). Fibre

d). Palmittic acid

[1x20 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

21. Name the major modern coconut processing technologies. [K] / [CO1]

22. List the major post-harvest causes of quality deterioration in coconut. [K] / [CO1]

23. State the important characteristics required for a suitable solvent used in coconut oil extraction. [K] / [CO2]

24. Explain the factors that influence oil recovery during cold pressing of coconut kernel. [U] / [CO2]

25. Explain the important quality parameters used for evaluating coconut flour. [U] / [CO3]

26. Write a note on expeller pressing method. [U] / [CO2]

27. State the importance of fat content, viscosity and emulsion stability in assessing the quality of coconut cream. [U] / [CO3]

28. Compare the processing sequence of wet and fresh-dry coconut flour production. [An] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

29. Compare coconut honey and coconut jam with respect to raw materials, major processing operations and final product characteristics. [An] / [CO4]

30. Examine the complete processing sequence of coconut cream production from mature coconut selection to final homogenization. [An] / [CO4]

31. Discuss the major practices involved in quality assurance of coconut products. [U] / [CO3]

32. Examine the complete processing sequence of coconut milk from selection of mature coconuts to packaging. [An] / [CO4]

[10x2 = 20]